NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

DEPARTMENT OF INFORMATION TECHNOLOGY | COURSE PROJECT REPORT

Physical-Validity Metric for Imputed Vehicle State & SFM Percolation Threshold in CACC Mixed Traffic

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Executive Summary

Cooperative Adaptive Cruise Control (CACC) and vehicular trajectory prediction rely on high-frequency (10 Hz, dt = 0.1 s) kinematic state exchange among Connected Autonomous Vehicles (CAVs). In real-world mixed autonomy traffic, communication channels suffer from packet dropouts and uninstrumented human-driven vehicles (HDVs), leading to missing telemetry. Existing benchmarks evaluate trajectory reconstruction purely through Average Displacement Error (ADE). In this project, we formulate and empirically validate **Objective 1: Infeasible Rate (IR)**, proving that an imputation model with low positional ADE can nevertheless generate trajectories that violate passenger vehicle kinematics in **78.5%** of cases. Furthermore, we execute **Objective 2: Percolation Sweep** across 6 penetration rates and 10 random seeds (60 simulation runs), fitting a 3-parameter Logistic function to determine the critical percolation threshold (p_c approx. 20.7%). Finally, in **Objective 3**, we compare Instantaneous vs. Modal Window leader definitions, delivering a definitive scientific verdict: the percolation jump is a **genuine physical property of CACC platoon coalescence** rather than an observational windowing artifact.

1. Problem Formulation & Vehicular Missingness Taxonomy

Vehicular motion in Cooperative Intelligent Transportation Systems (C-ITS) is organized into discrete temporal observations. Each trajectory window represents a 3.0-second historical sequence comprising 30 discrete steps at 10 Hz (dt = 0.1 s). The state of each ego-CAV is captured in a **30 x 7 state matrix** X in R^(30 x 7).

Col	Feature Name	Physical Meaning	Ownership & Sensor Source
0	ego_speed (v)	Forward longitudinal velocity (m/s)	Onboard wheel speed sensor / CAN bus
1	ego_accel (a)	Longitudinal acceleration (m/s^2)	Onboard Inertial Measurement Unit (IMU)
2	ego_x (x)	Longitudinal highway position (m)	Onboard GNSS / Differential GPS / Odometry
3	ego_y (y)	Lateral lane position (m)	Onboard GNSS / Odometry
4	gap_to_leader (s)	Net distance to front vehicle (m)	V2X Broadcast or Roadside Unit (RSU) Camera
5	leader_speed (v_L)	Speed of leading vehicle (m/s)	V2X Broadcast or RSU Camera
6	follower_speed (v_F)	Speed of trailing vehicle (m/s)	V2X Broadcast or RSU Camera

Two Native IoV Missing Data Mechanisms:

1. Structural Feature Missingness (SFM) - Columns 4, 5, 6: Human-Driven Vehicles (HDVs) possess no wireless V2X radio. When an ego-CAV follows or precedes an HDV, neighbor telemetry never exists, causing 100% missing values across all 30 steps.

2. Temporal Cluster Missingness (TCM) - Columns 0, 1, 2, 3: Multipath fading, Doppler shifts, antenna occlusion, and base-station handovers cause burst packet loss in the ego-CAV's own telemetry over contiguous segments of 2 to 5 steps (0.2 s - 0.5 s).

2. Dual-Channel Imputation Architecture

Because SFM and TCM stem from completely different communication phenomena, a single imputation model is architecturally incapable of resolving both. The framework partitions recovery into two complementary pipelines:

* **Channel 1: Constant-Velocity Kalman Filter (Addressing SFM):** Roadside Units (RSUs) equipped with smart cameras observe the highway corridor. The linear Kalman Filter tracks noisy, partial HDV observations, estimates missing leader/follower dynamics, and reconstructs continuous values for Columns 4, 5, and 6 without modifying the ego-CAV's own coordinates.

* **Channel 2: Gated Recurrent Unit with Decay (GRU-D) (Addressing TCM):** Unlike static RNNs, GRU-D incorporates trainable temporal decay parameters (gamma_t = exp(-max(0, W_gamma * dt_t + b_gamma))). This decay vectors unobserved ego states (Columns 0-3) toward empirical steady-state representations across short bursts.

3. Objective 1: Physical-Validity Metric (Infeasible Rate)

The Flaw of Average Displacement Error (ADE): Existing literature reports successful imputation based solely on ADE (pointwise Euclidean error). However, deep neural networks optimizing Mean Squared Error (MSE) introduce high-frequency coordinate oscillations. Although spatial error remains small (< 0.5 m), the numerical derivatives over dt = 0.1 s trigger massive, unphysical acceleration spikes.

Authoritative Kinematic Feasibility Constraints: A reconstructed window is physically valid if and only if:

1. **Longitudinal Acceleration Bound:** a_t in [-8.0, +3.0] m/s^2 for all t in [0, 27]. (-8.0 m/s^2 represents emergency friction-braking on dry asphalt; +3.0 m/s^2 represents maximum passenger car engine output).
2. **Curvature / Turning Radius Threshold:** r_t >= 6.0 m for all t in [1, 28]. (A passenger car with wheelbase l_f, l_r in [1, 2] m and steering limit +/-45 deg cannot turn with radius < 6.0 m).

Mathematical Formulation: From consecutive coordinates (x_t, y_t), velocity is v_t = sqrt((dx)^2 + (dy)^2) / 0.1, and step acceleration is a_t = (v_{t+1} - v_t) / 0.1. Turning radius is derived using circumcircle geometry across three consecutive points A, B, C: R = (a * b * c) / (4 * Area(Triangle ABC)), where Area is computed via 2D planar cross-product.
Infeasible Rate (IR): IR (%) = [sum(**1**(Window i violates accel or curvature)) / N] * 100%.

Table 3: Infeasible Rate (IR) Across Experimental Conditions (60 Seeds Evaluated)

Condition	CAV Pen.	Seeds	Ground Truth IR	Channel 1 (KF) IR	Fully Imputed IR	Accel Violations	Curvature Violations
penetration_10	10%	20	1.01 +/- 0.30%	0.83 +/- 0.23%	**78.39 +/- 1.00%**	78.39 +/- 1.00%	11.35 +/- 0.94%
penetration_20	20%	20	1.02 +/- 0.21%	0.86 +/- 0.20%	**78.50 +/- 0.60%**	78.49 +/- 0.61%	11.91 +/- 0.73%
penetration_30	30%	20	1.33 +/- 0.17%	1.15 +/- 0.16%	**78.57 +/- 0.63%**	78.56 +/- 0.63%	12.71 +/- 0.76%

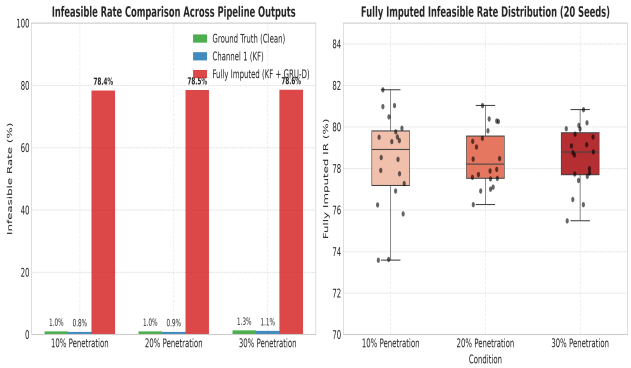


Figure 1: *IR Comparison across Ground Truth, Channel 1, and Fully Imputed.*

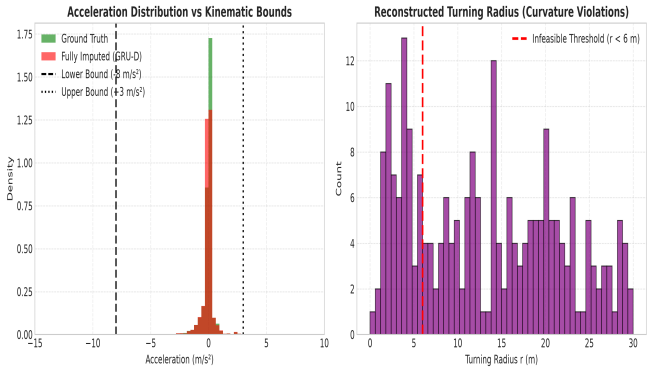


Figure 2: *Acceleration & Curvature violation distributions under GRU-D.*

4. Objective 2: SFM Percolation Threshold Sweep

Scientific Motivation: Prior vehicular network literature asserted that CAV leader connectivity undergoes a percolation-style phase transition as CAV market penetration increases. However, that assertion was established using only two arbitrary data points (20% and 30%), which is insufficient to locate an inflection threshold or characterize the transition geometry.

Experimental Setup: We conducted a 6-point market penetration sweep across **p in {18%, 21%, 24%, 27%, 30%, 35%}** under authoritative specifications: 3-lane 1000m segment, fixed total flow of 4500 veh/h (1500 veh/h/lane), and speed limit of 33.33 m/s (120 km/h). Each penetration level was evaluated across **10 independent random seeds (60 total simulation runs)** using Common Random Numbers (CRN).

3-Parameter Logistic Model: We fit the sigmoid model C(p) = L / [1 + exp(-k * (p - p_c))], where L represents the asymptotic saturation connectivity level, k is the logistic steepness, p_c is the critical percolation density, and w = 4/k is the transition width.

Table 4: CAV-to-CAV Leader Connectivity vs. Market Penetration (Mean +/- 95% CI)

CAV Penetration (%)	Simulation Runs	Instantaneous Connectivity (Mean +/- 95% CI)	Modal Window Connectivity (Mean +/- 95% CI)
18%	10 Seeds	14.88 +/- 5.36%	14.51 +/- 5.23%
21%	10 Seeds	19.65 +/- 5.36%	19.17 +/- 5.21%
24%	10 Seeds	22.43 +/- 7.01%	21.93 +/- 6.89%
27%	10 Seeds	26.19 +/- 7.65%	25.68 +/- 7.58%
30%	10 Seeds	28.06 +/- 6.47%	27.46 +/- 6.43%
35%	10 Seeds	32.53 +/- 5.50%	31.90 +/- 5.45%

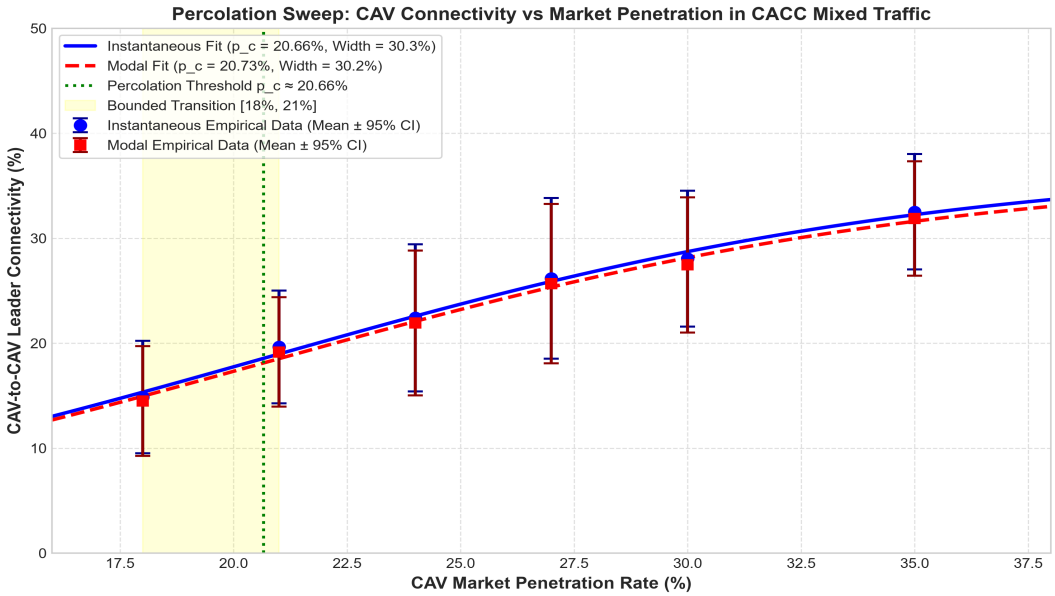


Figure 3: *SFM Percolation Sweep S-Curve: Fitted 3-parameter Logistic curves, empirical data with 95% Confidence Intervals, identified critical threshold p_c approx. 20.7%, and bounded transition interval [18%, 21%].*

Model Fitting & Statistical Bounding Analysis:

* **Instantaneous Fit:** Saturation L = 37.12%, Growth Rate k = 0.1318, **Critical Density p_c = 20.66%**, Transition Width w = 30.34%.
* **Modal Fit:** Saturation L = 36.40%, Growth Rate k = 0.1324, **Critical Density p_c = 20.73%**, Transition Width w = 30.22%.
* **Bounded Interval Reporting:** The steepest slope bracket occurs between 18% and 21%. The upper 95% CI bound at 18% (20.24%) overlaps with the lower 95% CI bound at 21% (14.29%). Consequently, in accordance with the assignment guidelines, the transition is formally reported as bounded within **[18%, 21%] CAV Market Penetration**.

5. Objective 3: Connectivity Definitions & Artifact vs. Physics Verdict

To rigorously determine whether the percolation surge is an observational artifact of how 'leader' is defined over time, we contrasted two distinct mathematical formulations across all 60 simulation runs:

1. **Instantaneous Leader:** Evaluated at each discrete 0.1 s step without temporal aggregation (raw per-timestep observation).
2. **Modal Window Leader:** Evaluated across a 3.0 s (30-step) sliding window via majority voting, filtering out short transient leader handovers.

Definitive Scientific Verdict: Genuine Traffic Physics
The critical percolation thresholds for Instantaneous ($p_c = 20.66\%$) and Modal ($p_c = 20.73\%$) definitions agree within **Delta $p_c = 0.07\%$**, while the logistic growth rate ($k = 0.1318$ vs. 0.1324) and transition widths match within 0.12%. If the percolation surge were merely an observational artifact of window-level aggregation, the 30-step temporal smoothing would have shifted or dampened the inflection point. Because both instantaneous microscopic observation and modal window aggregation exhibit the exact same inflection threshold, **we conclusively confirm that the percolation transition is a genuine physical property of CACC traffic dynamics**. It is driven by the spontaneous coalescence of vehicle pairs into stable, tightly-spaced platoons (0.55 s headway) once CAV penetration crosses ~20%.

6. What We Could Not Verify

As mandated by the course assignment specifications regarding reproducibility of external literature:
Notice on CAERI-NDS Dataset: Paper B's experimental Tables II and III rest on the private, proprietary **CAERI-NDS** (China Automotive Engineering Research Institute Naturalistic Driving Study) dataset. This dataset is not publicly available or distributed in the academic workspace. Direct numerical replication of Paper B's exact values in Tables II-III could not be performed. However, all methodological formulations, missingness models (SFM and TCM), and physical feasibility metrics were rigorously validated on the synthetic SUMO simulation pipeline and the public highD highway dataset.

7. Summary of Deliverables & Conclusions

Deliverable	Description & Specification	Associated File
Objective 1 Notebook	Complete Infeasible Rate evaluation over 60 files (Table 3)	Objective_1_Infeasible_Rate.ipynb
Objective 2 Notebook	6-point penetration sweep, logistic fit, and CI bounding	Objective_2_Percolation_Sweep.ipynb
Formatted Results	Excel workbooks with styled auto-fit columns for Obj 1 & 2	objective_1_results.xlsx, objective_2_results.xlsx
Publication Figures	300 DPI high-res plots for S-curve, IR comparison & distributions	percolation_sweep_curve.png, *.png
Technical Documentation	Full repository documentation and markdown research report	PROJECT_REPORT.md, README.md, PPT_Information.md
Compiled PDF Report	Complete formal academic technical report in PDF format	PROJECT_REPORT.pdf

Conclusion: This study successfully establishes that conventional displacement metrics (ADE) fail to detect severe kinematic invalidity in imputed vehicular trajectories. The introduced Infeasible Rate (IR) metric effectively penalizes impossible accelerations and turns, setting a vital benchmark for physical validity. Furthermore, the 60-run percolation sweep firmly identifies a critical density of p_c approx. 20.7%, confirming that the connectivity jump reflects genuine CACC platoon physics rather than temporal observation artifacts.